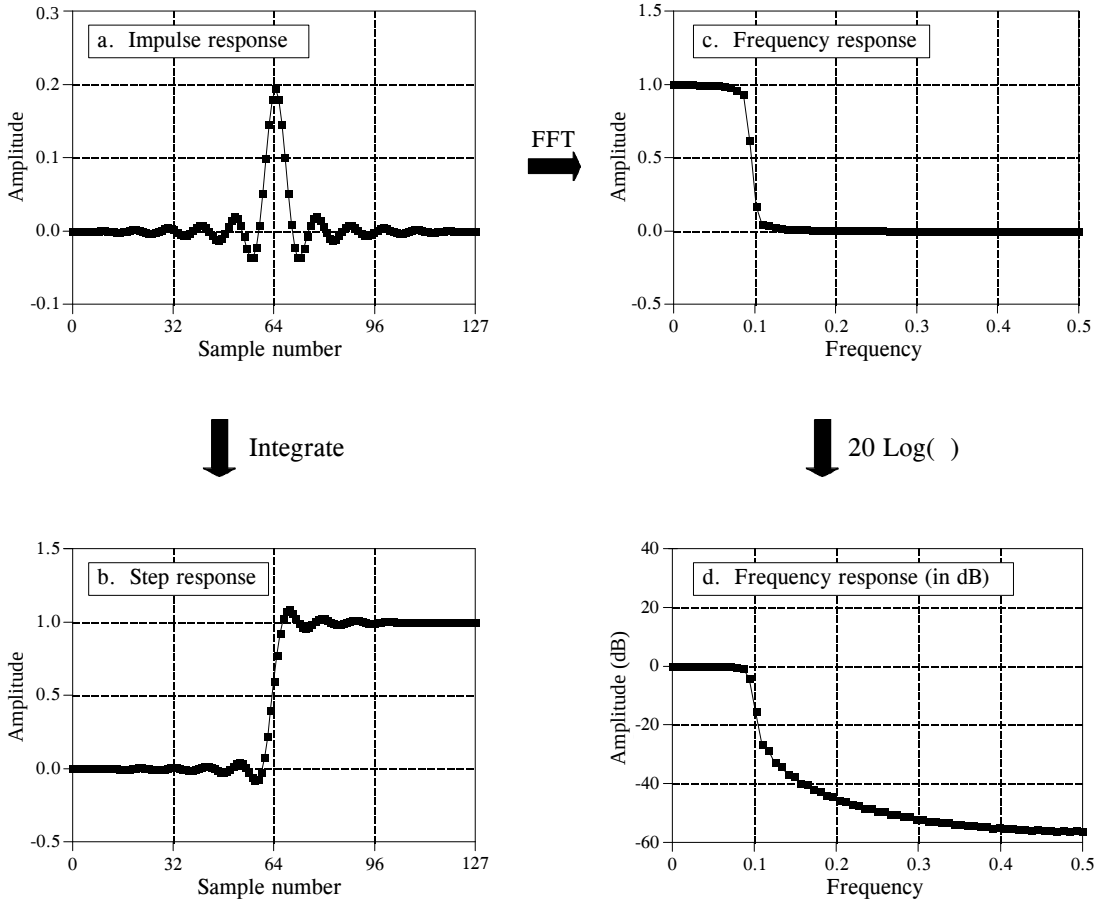

FIGURE 14-1
Filter parameters. Every linear filter has an impulse response, a step response, and a frequency response. The step response, (b), can be found by discrete integration of the impulse response, (a). The frequency response can be found from the impulse response by using the Fast Fourier Transform (FFT), and can be displayed either on a linear scale, (c), or in decibels, (d).

of this characteristic, recursive filters are also called **Infinite Impulse Response or IIR** filters. In comparison, filters carried out by convolution are called **Finite Impulse Response or FIR** filters.

As you know, the *impulse response* is the output of a system when the input is an *impulse*. In this same manner, the *step response* is the output when the input is a *step* (also called an *edge*, and an *edge response*). Since the step is the integral of the impulse, the step response is the integral of the impulse response. This provides two ways to find the step response: (1) feed a step waveform into the filter and see what comes out, or (2) integrate the impulse response. (To be mathematically correct: *integration* is used with continuous signals, while *discrete integration*, i.e., a running sum, is used with discrete signals). The frequency response can be found by taking the DFT (using the FFT algorithm) of the impulse response. This will be reviewed later in this